Pace to pass

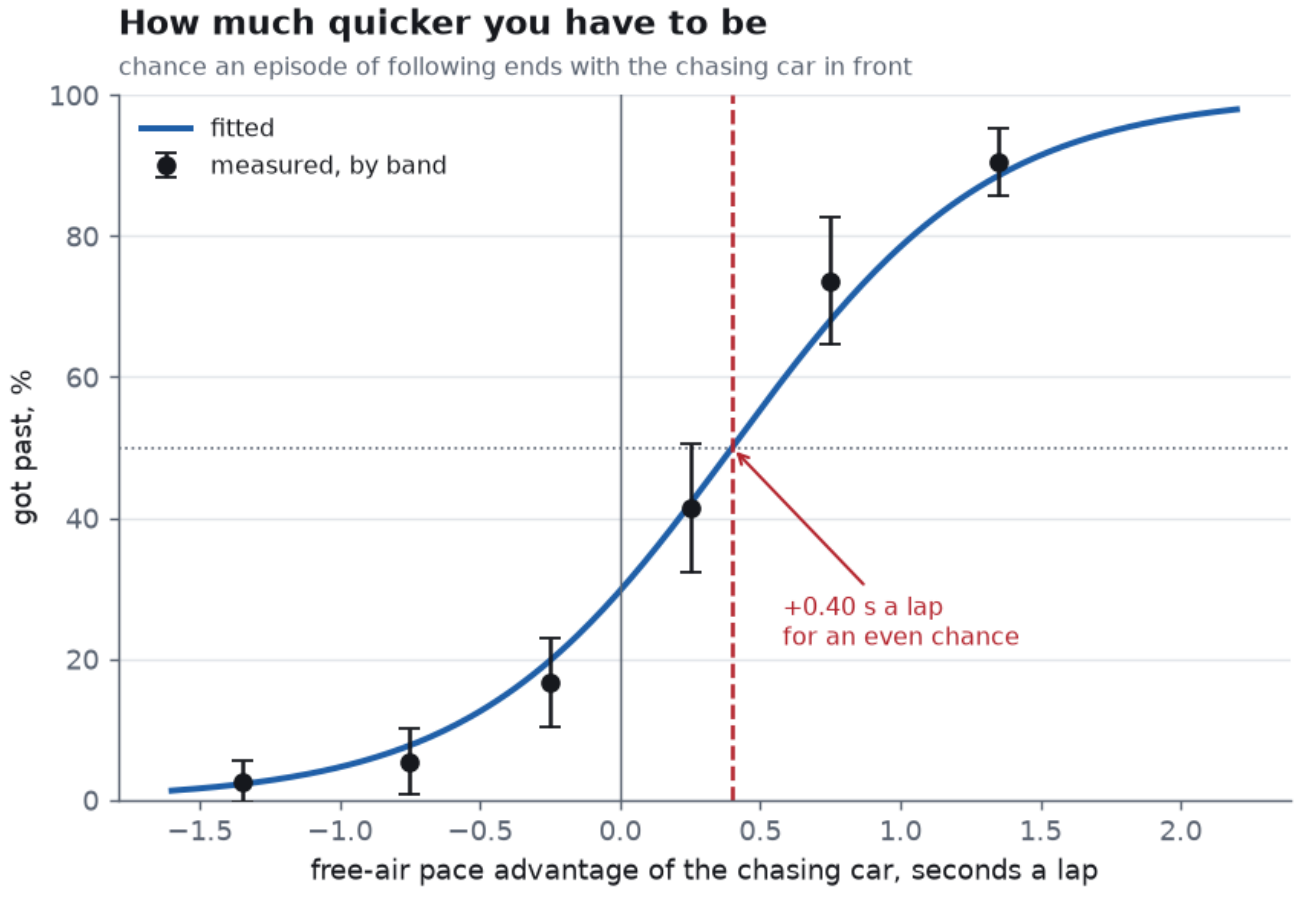
How much quicker a car has to be to get past the one in front, measured from 2023 lap times alone — and whether the circuit makes any difference.

It takes +0.40 seconds a lap to get past

At the mean opening gap of 1.24 s, over 689 episodes. Being held up costs +0.25 s a lap, and the circuit makes no measurable difference (p 0.47).

+0.40 s	689	+0.25 s	p 0.47
pace needed to get past	episodes measured	cost of being held up	circuit differences

The race order is in the lap times. Summing a car's lap times gives its elapsed race time, so sorting cars by that total at lap n puts them in the order they crossed the line. Against the position the feed reports, this agrees on 23,956 of 23,956 laps — 100.00%. Nothing here needs a positional feed, a gap feed or telemetry.



Which pairs are measured

outcome	episodes
pairs within reach	10,515
dropped: a pit lap	1,619
kept: never got past	1,574
kept: ended in a pass	869
dropped: the race ended first	92
dropped: a car stopped running	34

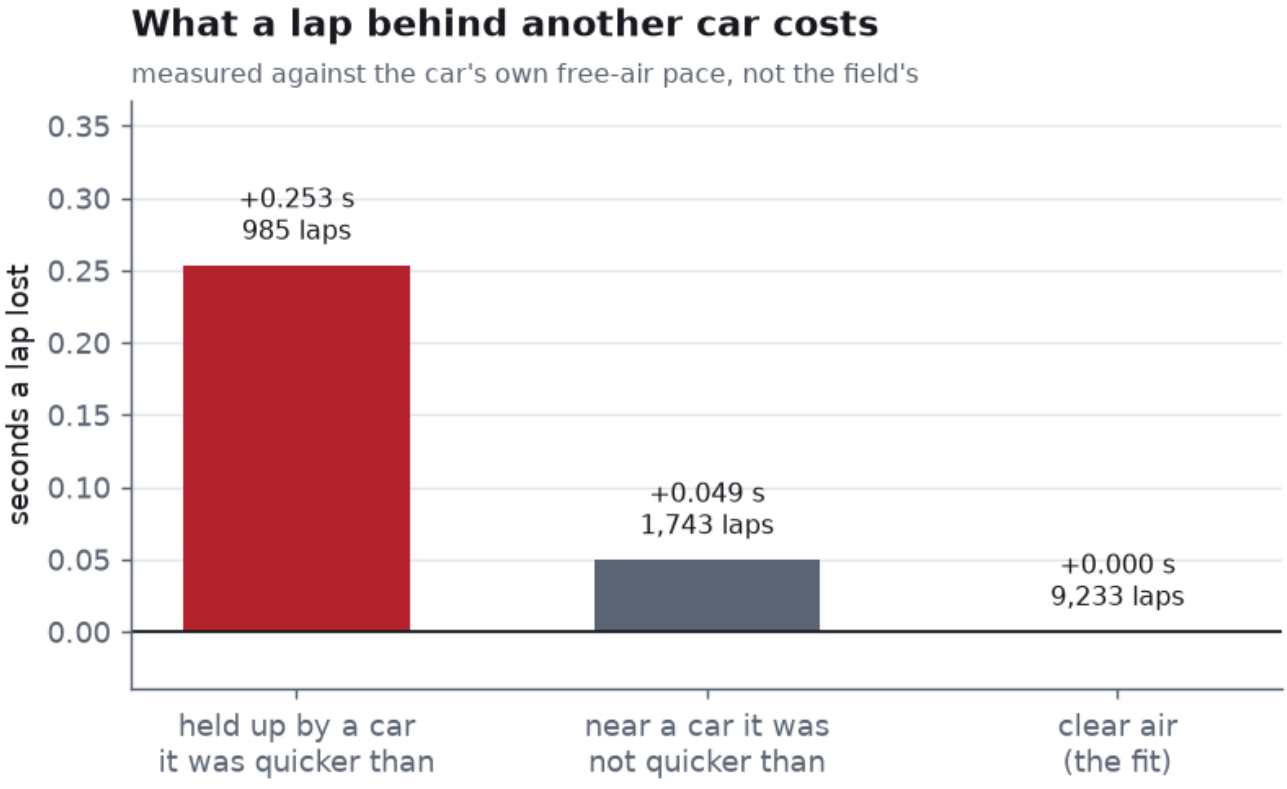
How quick each car was

The dose cannot be read off the laps two cars spend together, because those laps are the outcome: a car stuck behind another is slowed by being stuck behind it. So pace is measured apart and tested together — each car-race gets a line fitted to the laps it ran in clear air, and the advantage is the difference of two lines, each read at that car's own tyre age. The fitted field spans 2.57 s a lap at the median race, which is what a Formula 1 grid covers.

outcome	laps or car-races
clean laps	18,769
held out: in traffic	8,193
fitted car-races	295
refused: age and lap not separable	57
refused: too few clear laps	25

What being held up costs

The second bar is the control. Being close to another car costs nothing in itself — +0.049 s a lap for cars that were not quicker anyway. The cost appears only where the car behind had the pace to go faster and could not: +0.253 s a lap. Pooling the two, which is the obvious thing to do, dilutes it to almost nothing.



How much quicker you have to be

At the mean opening gap of 1.24 s, the fitted curve crosses even odds at +0.40 seconds a lap. The intercept is +0.013: two cars of identical pace running nose to tail are a coin toss, which is what it ought to be and was not imposed.

free-air advantage	episodes	got past
-9.0 to -1.0	115	2.6% ± 2.9%
-1.0 to -0.5	92	5.4% ± 4.6%
-0.5 to +0.0	132	16.7% ± 6.4%
+0.0 to +0.5	111	41.4% ± 9.2%
+0.5 to +1.0	91	73.6% ± 9.1%
+1.0 to +9.0	148	90.5% ± 4.7%

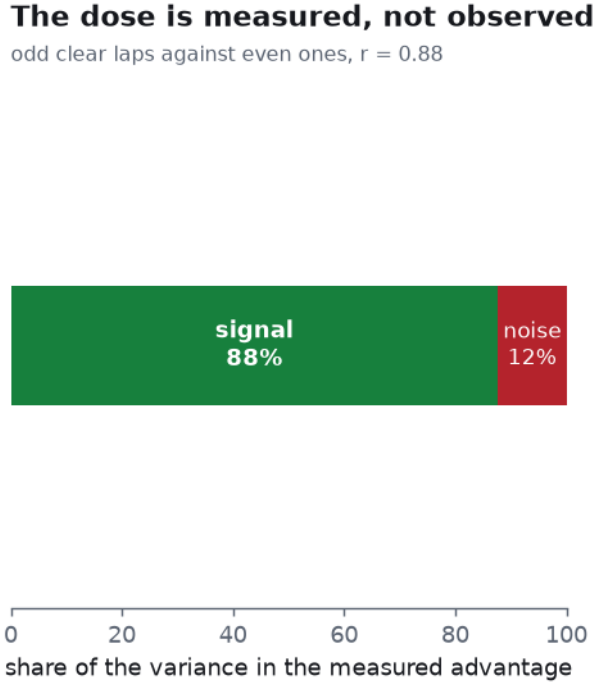
The fitted curve

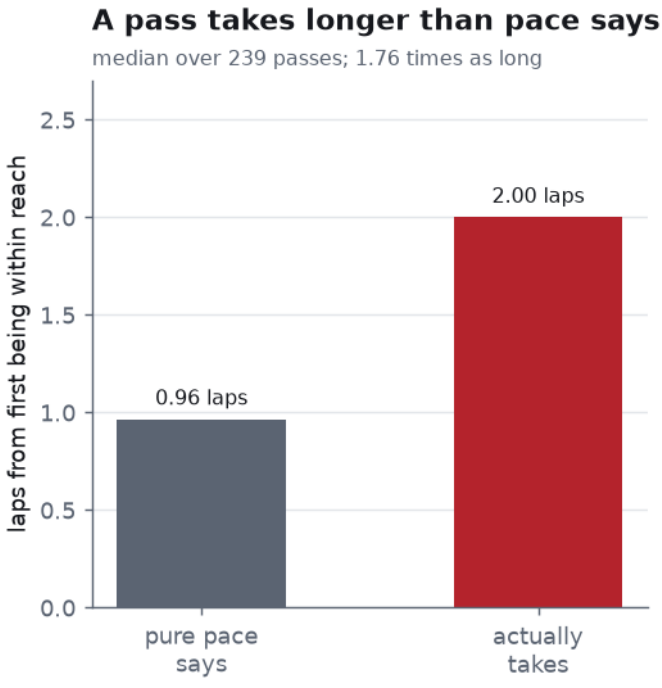
One logistic fit over every episode, with the opening gap as a control.

term	estimate	95% interval	p
base	+0.0125	-0.5287 to +0.5538	0.9637
per second per lap of advantage	+2.1552	+1.7314 to +2.5790	0.0000
per second of opening gap	-0.7067	-1.1044 to -0.3091	0.0005

The dose is measured, not observed

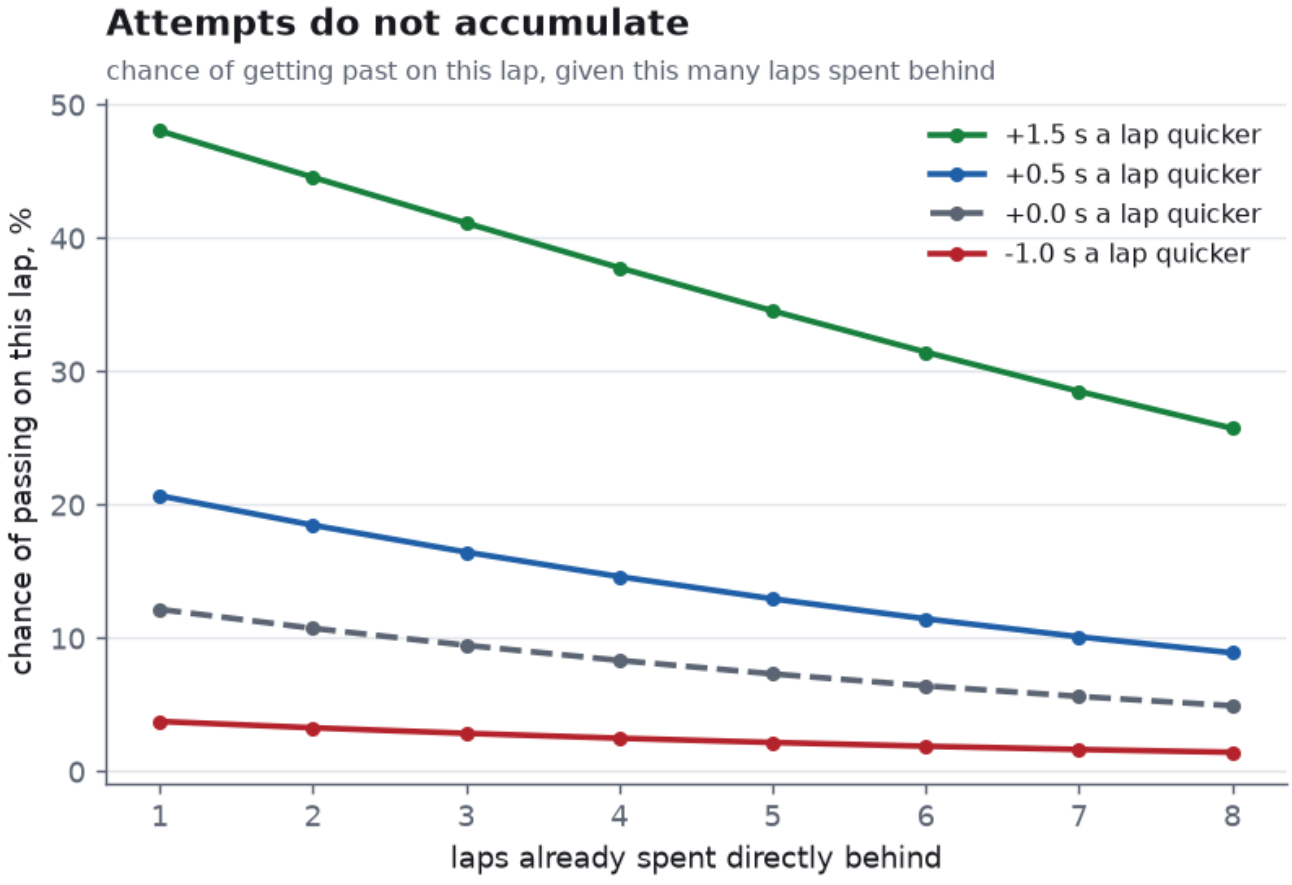
Fitting each car's line twice — once on its odd clear laps, once on its even ones — gives two independent measurements of the same quantity. They correlate 0.877, leaving noise of 0.376 s a lap against signal of 1.001. Reliability is 0.876, so the fitted slope is pulled toward zero by about that factor: the true slope is nearer 2.46 and the threshold nearer +0.35 s a lap.





Attempts do not accumulate

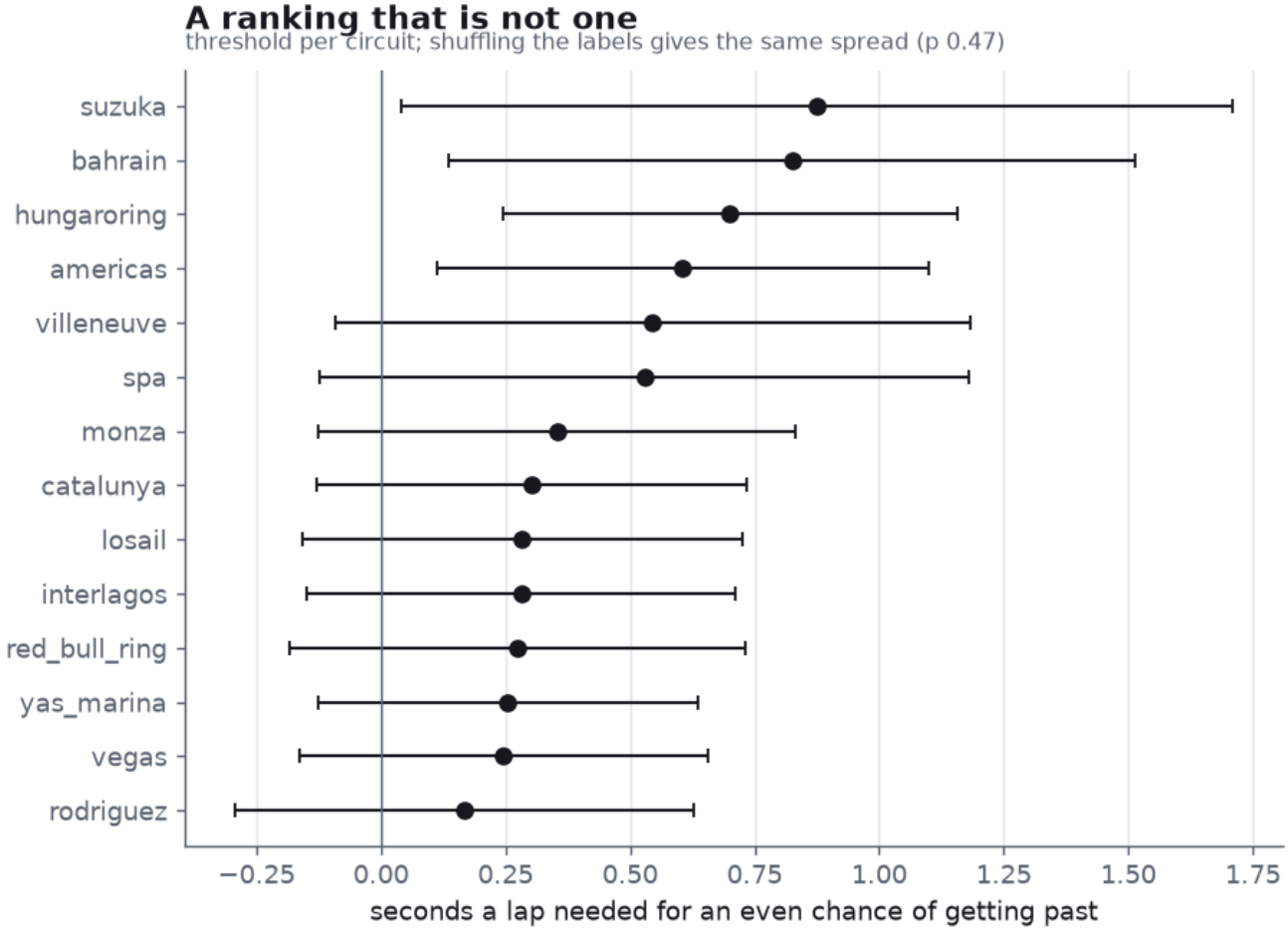
The time term is negative. A car that has not got past in the first couple of laps is less likely to manage it on the next one, not more. And a pass takes 1.76 times as long as pure pace says it should — 2.00 laps against 0.96 — which understates the resistance, because 33 more cars were still behind past the lap pace said they would be by and never got past at all.



term	estimate	95% interval	p
base	-0.8835	-1.3272 to -0.4399	0.0001
per second per lap of advantage	+1.2662	+0.9654 to +1.5669	0.0000
per lap of trying	-0.1402	-0.2498 to -0.0306	0.0123
per second of opening gap	-0.7691	-1.0491 to -0.4890	0.0000

Does the circuit matter? Not measurably

That column is not a ranking. Jointly, the 14 circuit terms give F 0.909, p 0.549. The fitted thresholds span 0.708 s a lap; shuffling the circuit labels across episodes — which destroys any real circuit effect and keeps everything else — gives a median span of 0.699, and beats the real spread 47% of the time. One season cannot resolve circuit differences, and the apparent ordering is what noise produces on its own.

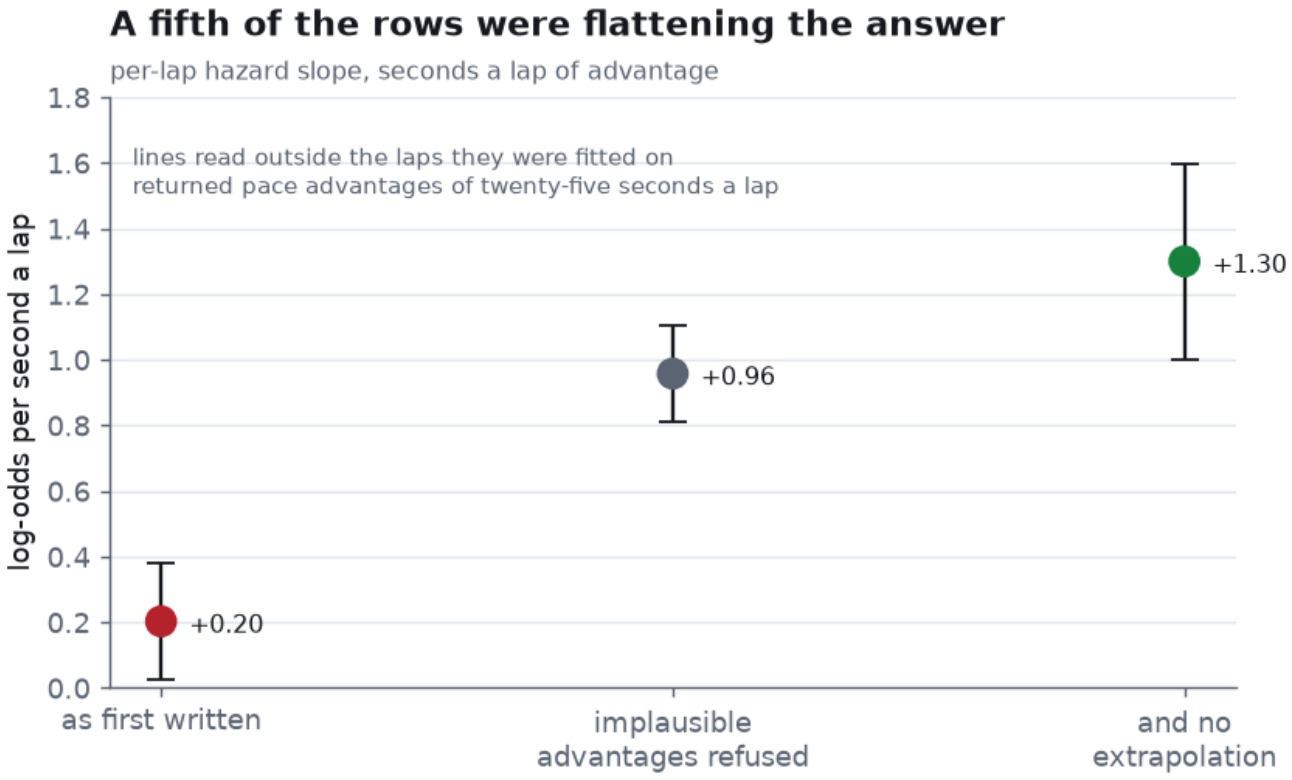


Does the circuit matter? Not measurably (continued)

circuit	threshold	95% interval	episodes	got past
suzuka	+0.87	+0.04 to +1.71	29	12
bahrain	+0.83	+0.14 to +1.51	31	9
hungaroring	+0.70	+0.25 to +1.16	43	16
americas	+0.61	+0.11 to +1.10	55	20
villeneuve	+0.54	-0.09 to +1.18	28	8
spa	+0.53	-0.12 to +1.18	39	16
monza	+0.35	-0.13 to +0.83	30	13
catalunya	+0.30	-0.13 to +0.73	52	27
losail	+0.28	-0.16 to +0.72	53	28
interlagos	+0.28	-0.15 to +0.71	35	13
red_bull_ring	+0.27	-0.18 to +0.73	49	17
yas_marina	+0.25	-0.13 to +0.63	90	39
vegas	+0.25	-0.16 to +0.66	65	27
rodriguez	+0.17	-0.29 to +0.63	33	14

What the guards were worth

A per-car line read outside the laps and tyre ages it was fitted on returned pace advantages of twenty-five seconds a lap. Two per cent of the rows were enough to flatten the central slope by a factor of five. Refusing implausible advantages recovers most of it; refusing extrapolation altogether recovers the rest, and the estimate then stops moving as the rule is tightened further.



Lap times and pit stops from the Jolpica mirror of the Ergast schema. The cleaning rules are carried over from the stint-degradation, race-strategy-mc and undercut-effect projects.